

SANJIVANI UNIVERSITY
School of Engineering and Technology

AGILE MINI-PROJECT – UNIT 2

Student Record and Academic Management Platform

Submitted By

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1. Product Vision

The Student Record and Academic Management Platform is planned as a web-based system for colleges to manage student and academic information in one place. The main aim is to reduce manual record keeping and make common academic activities faster and easier for students, faculty members and administrators.

Product Vision: Develop a secure and user-friendly platform that helps educational institutions manage student records, attendance, marks, courses and academic reports through a single web application.

Expected Outcomes

- Student information can be stored and updated easily.
- Faculty can record attendance and marks without maintaining separate registers.
- Students can view their academic information from one place.
- Authorized users can generate useful academic reports.
- The system keeps access controlled according to user roles.

Success Criteria

Measure	Target
Student record management	Add, search, update and delete records successfully
Attendance	Attendance can be recorded and viewed for a selected course/date
Marks	Marks can be entered and grades/results calculated correctly
Security	Only authorized users can access restricted modules
Usability	A new user should be able to complete common tasks with minimal guidance

2. Primary Users and Stakeholders

User / Stakeholder	Main Need
Administrator	Manage users, student records, courses and reports
Faculty Member	Manage courses, attendance and marks
Student	View profile, attendance, courses and results
Department / Management	Access academic summaries and reports
Project Team	Develop, test and maintain the application

3. Scrum Team Structure

Role	Assigned Responsibility
Product Owner	Maintains requirements, prioritizes backlog and represents stakeholder needs
Scrum Master	Conducts Scrum meetings, tracks impediments and supports the team
Development Team	Designs, codes, tests and documents the application

Team member names can be added after the group finalizes the allocation of roles.

4. Scrum Roles and Responsibilities

Product Owner

- Collect requirements from stakeholders.
- Prepare and prioritize the Product Backlog.
- Clarify user stories for the development team.
- Review completed work and accept or request changes.

Scrum Master

- Schedule and facilitate Sprint Planning, Daily Scrum, Review and Retrospective.
- Help the team remove blockers.
- Encourage proper use of Scrum practices.
- Keep communication within the team clear.

Development Team

- Design the user interface and database.
- Implement application features.
- Perform unit and functional testing.
- Maintain project documentation and demonstrate completed work.

5. Project Epics

Epic	Purpose	Major Features
E1 – Student Management	Maintain complete student information	Registration, profile, search, update, delete
E2 – Course Management	Manage courses and	Add course, update course,

E3 – Attendance Management	student-course assignment	assign students, course details
E4 – Marks & Academic Performance	Manage marks and academic results	Enter marks, update marks, calculate grades, result sheets
E5 – Authentication & Security	Protect the system and control access	Login, logout, password reset, role-based access
E6 – Notifications	Inform users about important academic events	Attendance, examination and result notifications
E7 – Reporting	Provide useful academic summaries	Student, attendance, performance and course reports

6. User Stories

ID	Epic	User Story
US01	Authentication	As a registered user, I want to log in with my credentials so that I can access the features available to my role.
US02	Authentication	As a user, I want to log out securely so that my account is not left open on a shared device.
US03	Student Management	As an administrator, I want to add a student record so that new students can be maintained in the system.
US04	Student Management	As an administrator, I want to search for a student so that I can quickly find the required record.
US05	Student Management	As an administrator, I want to update student details so that records remain current.
US06	Student Management	As an administrator, I want to delete an incorrect or inactive student record so that the database remains accurate.
US07	Course Management	As an administrator, I want to add a course so that it is available for academic

		activities.
US08	Course Management	As an administrator, I want to assign students to a course so that course-wise records can be maintained.
US09	Attendance	As a faculty member, I want to record attendance for a selected course and date so that attendance remains accurate.
US10	Attendance	As a faculty member, I want to view attendance reports so that I can identify attendance shortages.
US11	Marks	As a faculty member, I want to enter student marks so that academic performance is recorded.
US12	Marks	As a student, I want to view my result and grades so that I can check my academic performance.
US13	Notifications	As a student, I want to receive important academic notifications so that I do not miss examination or result updates.
US14	Reporting	As an administrator, I want to generate student and academic reports so that management can review the data.

7. Acceptance Criteria

Story	Acceptance Criteria
US01	Login page opens; registered user can enter credentials; valid credentials open the dashboard; invalid credentials show an error.
US03	Required student fields are validated; a valid record is saved; duplicate/invalid data is handled with a suitable message.
US04	Search accepts a relevant student identifier/

US05	name; matching records are displayed; no-match condition is shown clearly.
US07	Course details can be entered; required fields are validated; valid course is saved and displayed in the course list.
US08	A course can be selected; eligible students can be assigned; assignment is saved and can be viewed later.
US09	Faculty can select course and date; attendance can be marked; record is saved successfully; duplicate entry is prevented/handled.
US10	Faculty can select course/date or relevant filters; attendance data is displayed; report can be generated.
US11	Faculty can select a student/course; marks are validated; valid marks are saved; updated marks are reflected in the record.
US12	Student can access own result; marks and grades are displayed correctly; unauthorized results are not shown.
US13	Notification contains relevant academic information; intended users can view it; old notifications remain accessible when required.
US14	Authorized user can select report type; report contains correct data; report can be viewed/exported.

8. Prioritized Product Backlog

ID	Story	Epic	Priority	Points
US01	User Login	Authentication	High	5
US03	Add Student	Student Management	High	3
US04	Search Student	Student Management	High	3
US09	Record Attendance	Attendance	High	5
US07	Add Course	Course Management	High	3

US05	Update Student	Student Management	Medium	3
US11	Enter Marks	Marks	Medium	5
US10	Attendance Report	Reporting	Medium	5
US12	View Result	Marks	Medium	3
US02	Logout	Authentication	Medium	2
US14	Generate Academic Reports	Reporting	Low	8
US13	Notifications	Notifications	Low	5
US06	Delete Student	Student Management	Low	3

9. Story Point Estimation

Points	Complexity	Typical Meaning
1	Very Easy	Small change with almost no dependency
2	Easy	Simple feature with limited work
3	Moderate	Normal feature with some validation/testing
5	Difficult	Multiple screens, logic or dependencies
8	Very Difficult	Large feature needing considerable design/testing
13	Highly Complex	Should normally be split into smaller stories

The team uses Planning Poker style discussion. The estimate represents relative effort, not exact development hours.

10. Sprint Planning

Sprint	Goal	Planned Work	Expected Deliverable
Sprint 1	Set up the project and authentication	Repository setup, basic UI, login, logout, initial testing	Working login flow and project skeleton
Sprint 2	Develop Student Management	Add, search, update and delete student	Working student management

Sprint 3	Develop Course and Attendance	Add course, assign students, record attendance, attendance report	module
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11. Sprint Backlog

Sprint 1

Task	Status at Start
Initialize repository and project structure	To Do
Create basic UI	To Do
Implement login	To Do
Implement logout/session handling	To Do
Test authentication	To Do

Sprint 2

Task	Status at Start
Create student form	To Do
Implement add student	To Do
Implement search	To Do
Implement update	To Do
Implement delete	To Do
Test student module	To Do

Sprint 3

Task	Status at Start
Create course form	To Do
Implement course management	To Do
Assign students to courses	To Do
Create attendance screen	To Do
Implement attendance report	To Do
Test course and attendance	To Do

12. Scrum Board

Backlog	To Do	In Progress	Testing	Completed
Notifications	Add Student	Login	—	Repository Setup

Academic Reports	Search Student	—	—	—
Result Sheet	Add Course	—	—	—
Password Reset	Record Attendance	—	—	—

The board will be updated during sprint execution. Items move from To Do → In Progress → Testing → Completed.

13. Scrum Ceremonies

Sprint Planning

The team selects the highest-priority stories that can reasonably be completed in the sprint, discusses dependencies, estimates effort and agrees on the Sprint Goal.

Daily Scrum

Each member briefly reports what was completed, what will be done next and any blocker affecting the work.

Sprint Review

The team demonstrates completed features to the Product Owner/stakeholders and records feedback for the Product Backlog.

Sprint Retrospective

The team discusses what went well, what caused difficulties and one or two practical improvements for the next sprint.

14. Definition of Done

- Feature is implemented according to the user story.
- Acceptance criteria are satisfied.
- Basic testing is completed and major defects are fixed.
- Code is integrated into the project repository.
- Required documentation is updated.
- Feature is demonstrated during Sprint Review.

15. Agile Framework Comparison

Framework	Main Idea	Suitability for This Project
Scrum	Time-boxed sprints, defined roles and regular reviews	High – suitable for a small student team with changing requirements
Kanban	Continuous flow using a	Good – useful when tasks

	visual board and work-in-progress limits	arrive continuously
XP	Strong focus on engineering practices, testing and frequent releases	Good – useful for code quality, especially with a small development team
Lean Development	Remove waste and deliver customer value quickly	Good – useful as a supporting approach
LeSS	Scales Scrum across multiple teams with a shared product	Low – more than needed for this project size
SAFe	Large-scale framework for coordinating many teams and business levels	Low – too heavy for a student project
Scrum@Scale	Coordinates multiple Scrum teams using scaling patterns	Low – unnecessary for a single small team

16. Recommended Framework

Scrum is recommended for the Student Record and Academic Management Platform. The project can be divided into short sprints, and the team can demonstrate working features after each sprint. The defined Product Owner, Scrum Master and Development Team roles also make responsibilities clear. For a student project with a limited team size, Scrum gives enough structure without adding the complexity of large-scale frameworks.

17. Project Timeline and Release Roadmap

Phase	Planned Work	Output
Phase 1	Requirements and backlog preparation	Vision, epics, stories and prioritized backlog
Sprint 1	Authentication and project setup	Initial working application
Sprint 2	Student Management	Student module
Sprint 3	Course and Attendance	Course + attendance modules
Next Release	Marks, notifications and reports	Expanded academic management platform
Final Release	Integration, testing and documentation	Complete project demonstration

18. Burndown Chart Data

The following sample data can be used to plot the sprint burndown chart after the team starts tracking actual progress.

Sprint Day	Planned Remaining Points	Example Actual Remaining Points
Day 1	26	26
Day 2	21	23
Day 3	16	19
Day 4	11	14
Day 5	6	9
Day 6	0	3
Day 7	0	0

19. Sprint Review Report – Format

Item	Details to Record
Sprint Goal	Whether the sprint goal was achieved
Completed Stories	Stories demonstrated successfully
Not Completed	Stories carried forward and reason
Feedback	Suggestions received from Product Owner/ users
Action	Backlog changes or improvements based on feedback

20. Sprint Retrospective Report – Format

Question	Team Notes
What went well?	Good communication / completed planned work / useful testing
What was difficult?	Dependencies, debugging, unclear requirements, time management
What should improve?	Smaller tasks, earlier testing, clearer task ownership
Action for next sprint	Agree on one or two measurable improvements

21. Conclusion

This Agile plan converts the Student Record and Academic Management Platform requirements into manageable epics, user stories and sprints. Prioritizing the backlog helps the team work on the most useful features first. Regular Scrum meetings, testing and reviews provide a simple way to track progress and improve the product throughout development.